

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459365.7895 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1281 +/- 0.0092
Transit depth (Rp/Rs)^2	1.64 +/- 0.24 [%]
Orbital Inclination (inc)	86.62 +/- 0.57
Airmass coefficient 1 (a1)	1.342 +/- 0.012
Airmass coefficient 2 (a2)	-0.2229 +/- 0.006
Scatter in the residuals of the lightcurve fit is	0.85 %
Best Comparison Star	None
Optimal Aperture	4.1
Optimal Annulus	9.57
Transit Duration (day)	0.0834 +/- 0.0063

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1281 ± 0.0092 vs 0.1063 (deviation 2.37σ)
Duration fit vs published	0.0834 d vs 0.0867 d (deviation 0.52σ)
Mid-time O–C	-7.4 min
Depth SNR	13.9
Residual scatter	0.85 %

Light curve

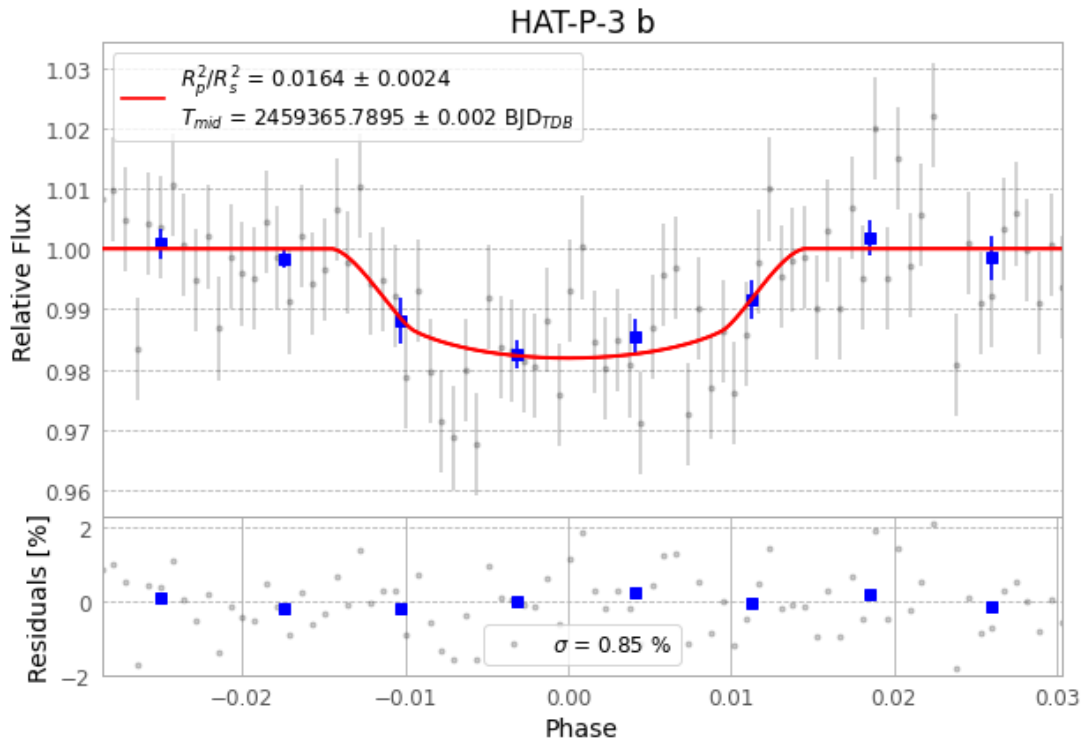


Figure 1 — FinalLightCurve_HAT-P-3 b_2021-05-30.png (SHA-256 4a9aac90d26f4209...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [408, 236], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d1a4d081af7fd8bb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

83 FITS frames (55 MB), HATP-3210531045414.FITS → HATP-3210531090012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-3-210531.log (16a0d26d29ae...), FinalLightCurve_HAT-P-3 b_2021-05-30.png (4a9aac90d26f...), FinalLightCurve_HAT-P-3 b_2021-05-30.csv (44b49d2f9566...)

Compute resources

Wall time 140.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 23:53Z.